Computer Organization & Architecture

Unit- 9
Memory Organization

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Terminology – Memory is important?

- In terms of Processor and Memory both are important.

1) Storage of data
2) Multitasking

Eg, Good cricketer means not bats man only same purpose bowler also a good cricketer

Terms of computer good Processor is batsman but memory is also good bowler.

Terminology – Why we required memory Hierarchy ?

- **If we have only one memory then we can done all things?**

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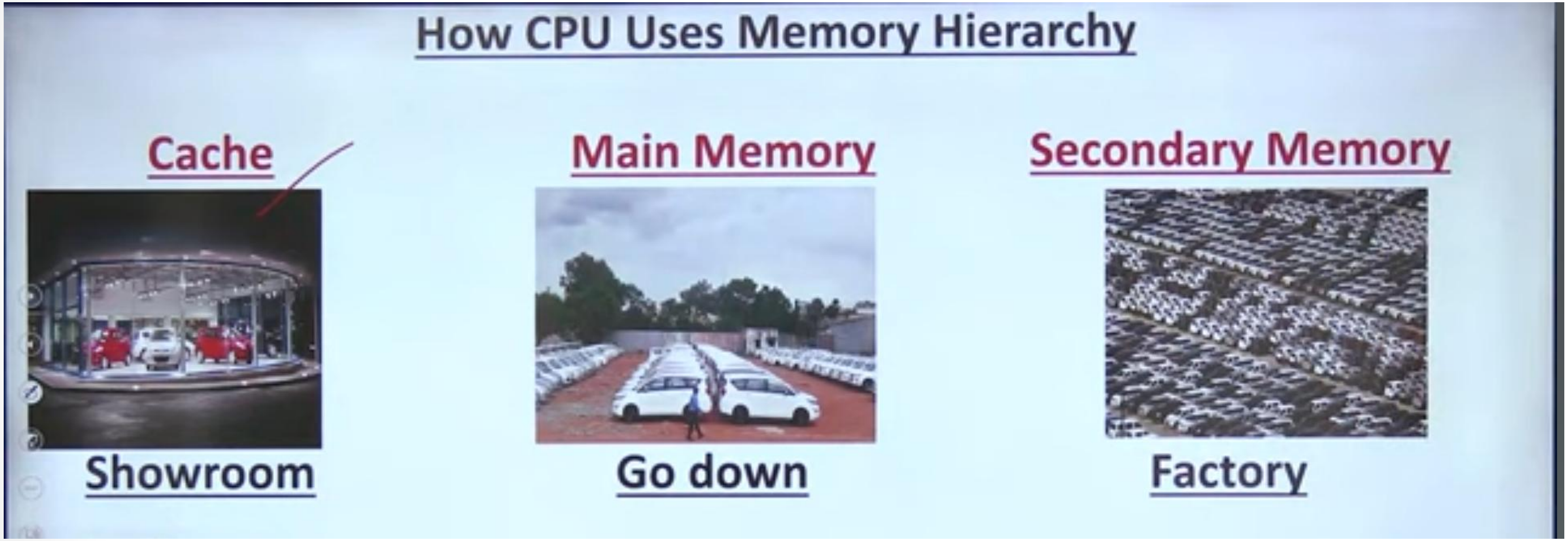
Terminology – How Computer use Memory Hierarchy



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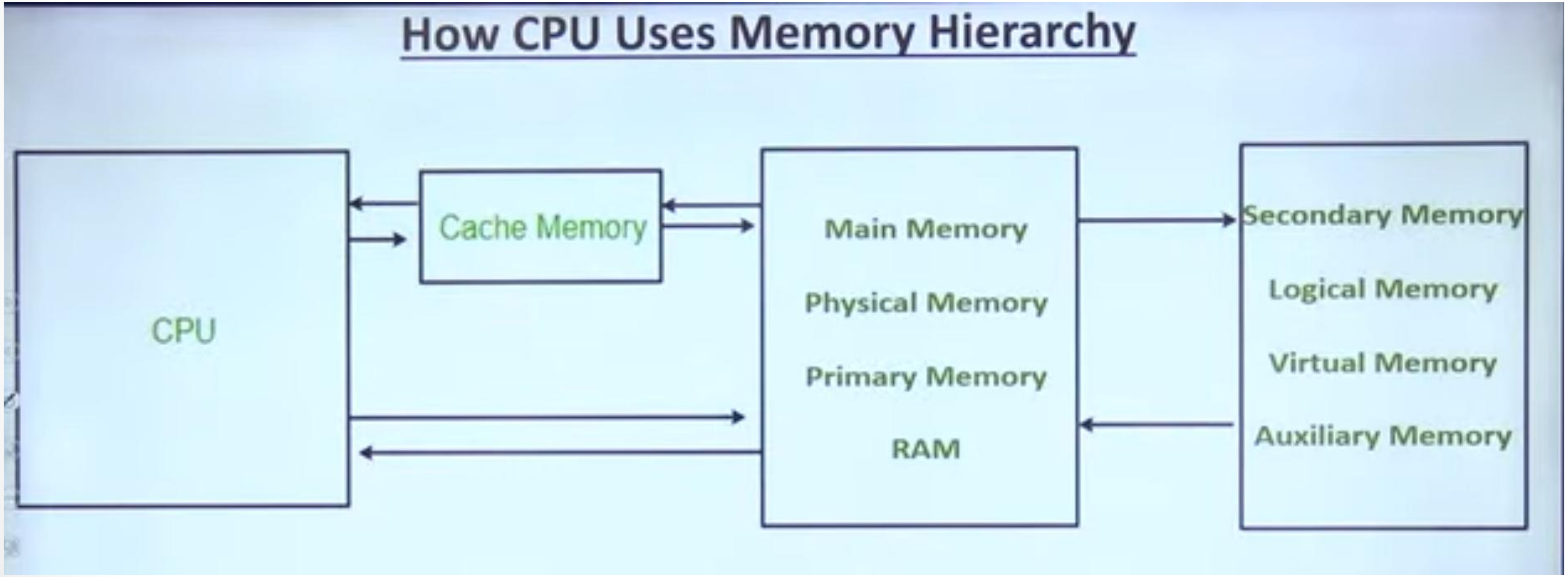
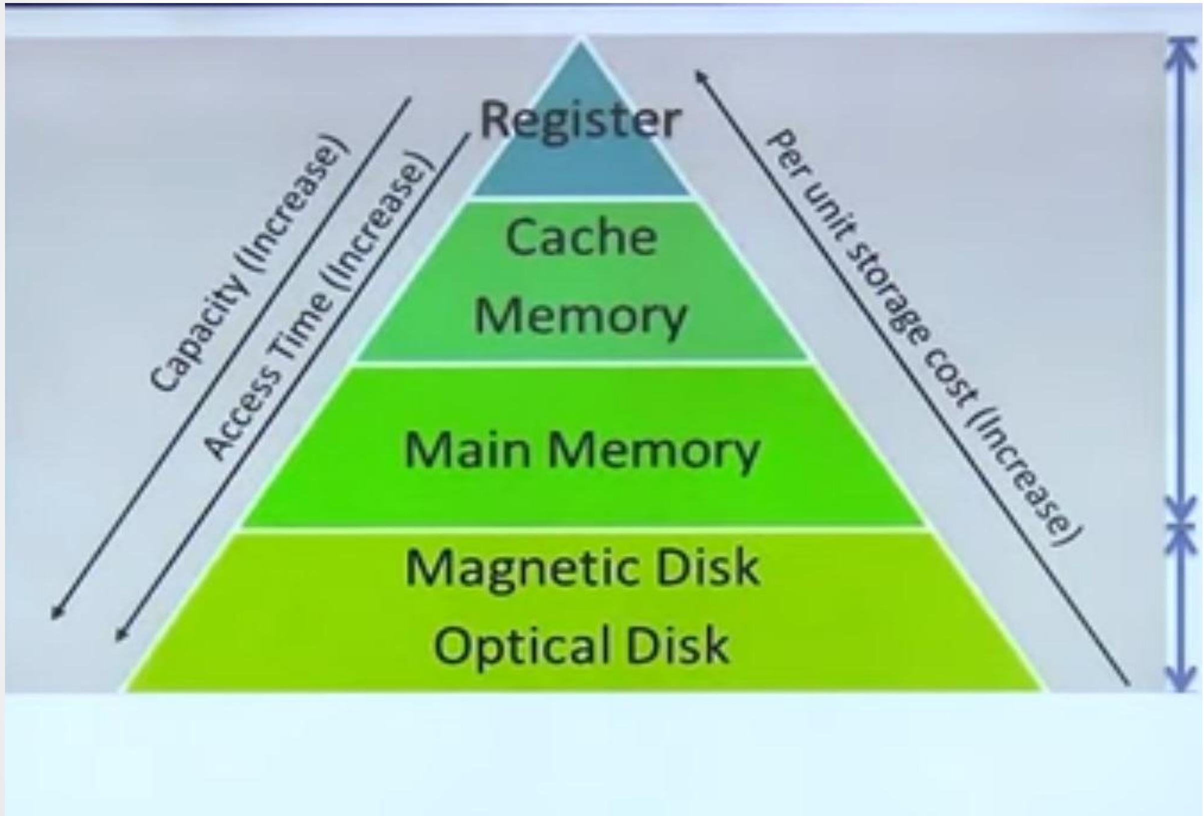
How Computer use Memory Hierarchy ?



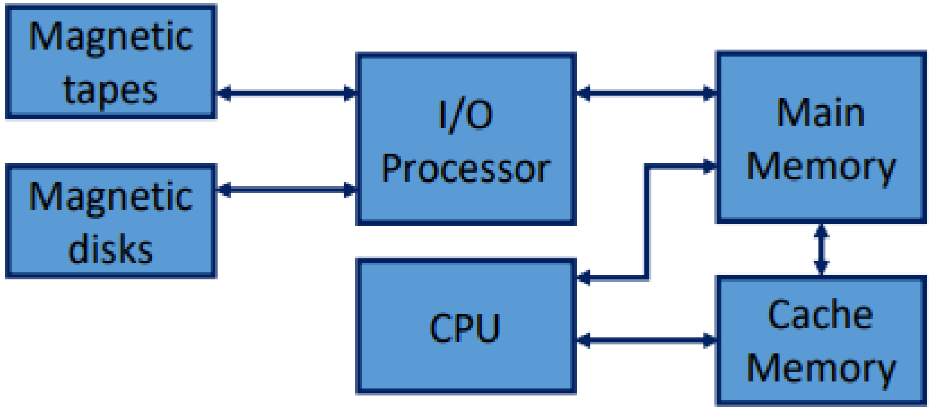
Memory Hierarchy



Memory Type	Total Cost	Capacity	Cost per GB
Magnetic Disk (HDD)	₹2,000	1000 GB	₹2 / GB
Main Memory (RAM)	₹4,000	8 GB	₹500 / GB
Cache Memory	₹1,000	0.25 GB	₹4,000 / GB
Register	₹500	0.0001 GB	₹5,000,000 / GB

Memory Hierarchy

- The memory unit is an essential component in any digital computer since it is needed for storing programs and data.
- The memory unit that communicates directly with the CPU is called the **main memory.** Devices that provide backup storage are called **auxiliary memory.**
- The most common auxiliary memory devices used in computer systems are magnetic disks and tapes. They are used for storing system programs, large data files, and other backup information.
- A special very-high speed memory called a **cache** is sometimes used to increase the speed of processing by making current programs and data available to the CPU at a rapid rate.
- The cache is used for storing segments of programs currently being executed in the CPU and temporary data frequently needed in the present calculations.



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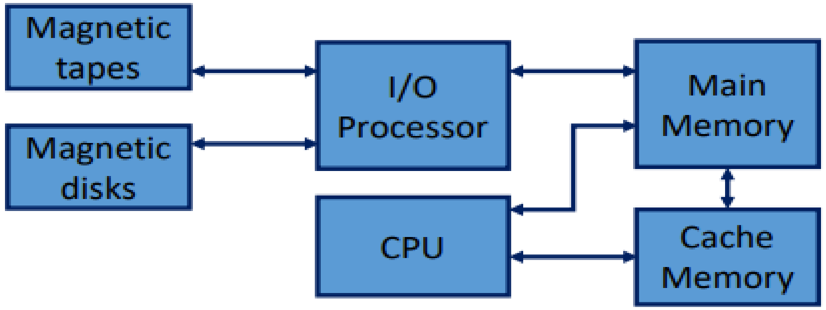
Memory Hierarchy



- While the I/O processor manages data transfers between auxiliary memory and main memory, the cache organization is concerned with the transfer of information between main memory and CPU.
- The reason for having two or three levels of memory hierarchy is economics.
- As the storage capacity of the memory increases, the cost per bit for storing binary information decreases and the access time of the memory becomes longer.
- **The auxiliary memory** has a large storage capacity, is relatively inexpensive, but has low access speed compared to main memory.
- **The cache memory** is very small, relatively expensive, and has very high access speed.
- Auxiliary and cache memories are used for different purposes. The cache holds those parts of the program and data that are most heavily used, while the auxiliary memory holds those parts that are not presently used by the CPU.

- Moreover, the CPU has direct access to both cache and main memory but not to auxiliary memory.

Eg , movie show after we scrolled and buffering

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Main Memory

- The main memory acts as the central storage unit in a computer system. It is a relatively large and fast memory which is used to store programs and data.
- The integrated circuits for the main memory are classified into two major units:

1. RAM (Random Access Memory)
2. ROM (Read Only Memory)

RAM(Random Access Memory)

- Used in computers for the temporary storage of programs and data.
- Read and write both operations are performed by RAM.
- It is volatile and lose all stored information if power is interrupted or turned off.
- RAMs typically come with word capacities of 1K, 4K, 8K, 16K, etc..
- RAM is also divided into two types – Static and Dynamic
- It can be expanded by combining several memory chips.

Main Memory

<u>ROM(Read only Memory)</u>

- ROM is used for storing programs that are permanently resident in the computer and for tables of constants that do not change in value once the production of the computer is completed.
- The ROM portion of main memory is **needed for storing an initial program** called a bootstrap loader. The bootstrap loader is a program whose function is to start the computer software operating when power is turned on.
- The bootstrap program loads a portion of the operating system from disk to main memory and control is then transferred to the operating system, which prepares the computer for general use.
- The **contents of ROM remain unchanged after power is** turned off and on again.

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RAM Chips

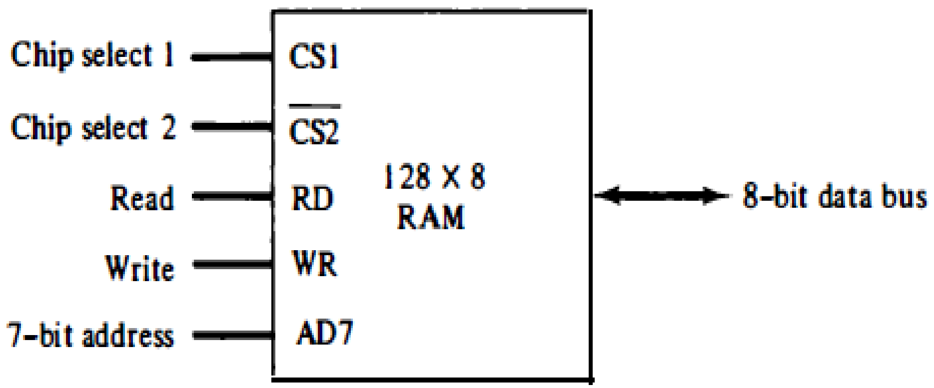
- The capacity of the memory is 128 words of eight bits (one byte) per word. This requires a 7-bit address and an 8-bit bidirectional data bus.

- The read and write inputs specify the memory operation and the two chips select (CS) control inputs are for enabling the chip only when it is selected by the microprocessor.

- When CS1 = 1(active high signal) and CS2 = 0 (active low signal) the memory can be placed in a write or read mode.

- If the chip select inputs are not enabled, or if they are enabled but the read or write inputs are not enabled, the memory is inhibited and its data bus is in a **high-impedance state**

Figure 12-2 Typical RAM chip.



(a) Block diagram

CS1	$\overline{CS2}$	RD	WR	Memory function	State of data bus
0	0	×	×	Inhibit	High-impedance
0	1	×	×	Inhibit	High-impedance
1	0	0	0	Inhibit	High-impedance
1	0	0	1	Write	Input data to RAM
1	0	1	×	Read	Output data from RAM
1	1	×	×	Inhibit	High-impedance

(b) Function table

ROM Chips

- However, since **a ROM can only read, the data bus can only be in an output mode.**
- The two chip select inputs must be CS1 = 1 and CS2 = 0 for the unit to operate. Otherwise, the data bus is in a high-impedance state.

- The nine address lines in the ROM chip.
- Thus when the chip is enabled by the two select inputs, the byte selected by the address lines transfer on the data bus.

- There is no need for a write control because the unit can only read.

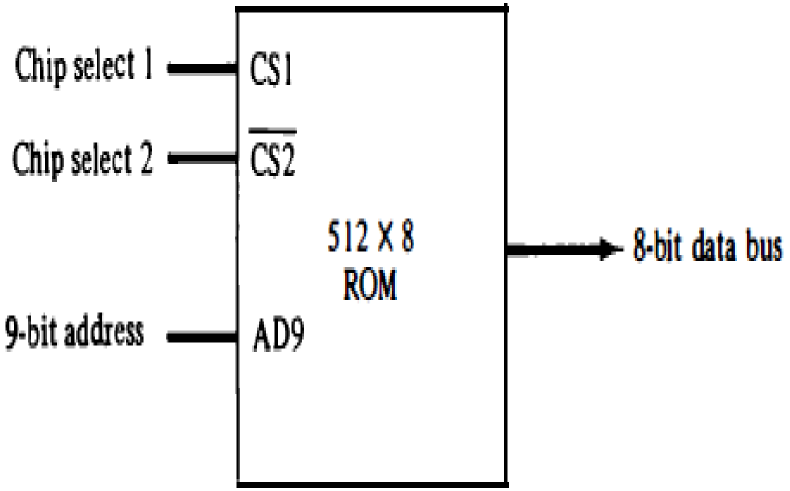
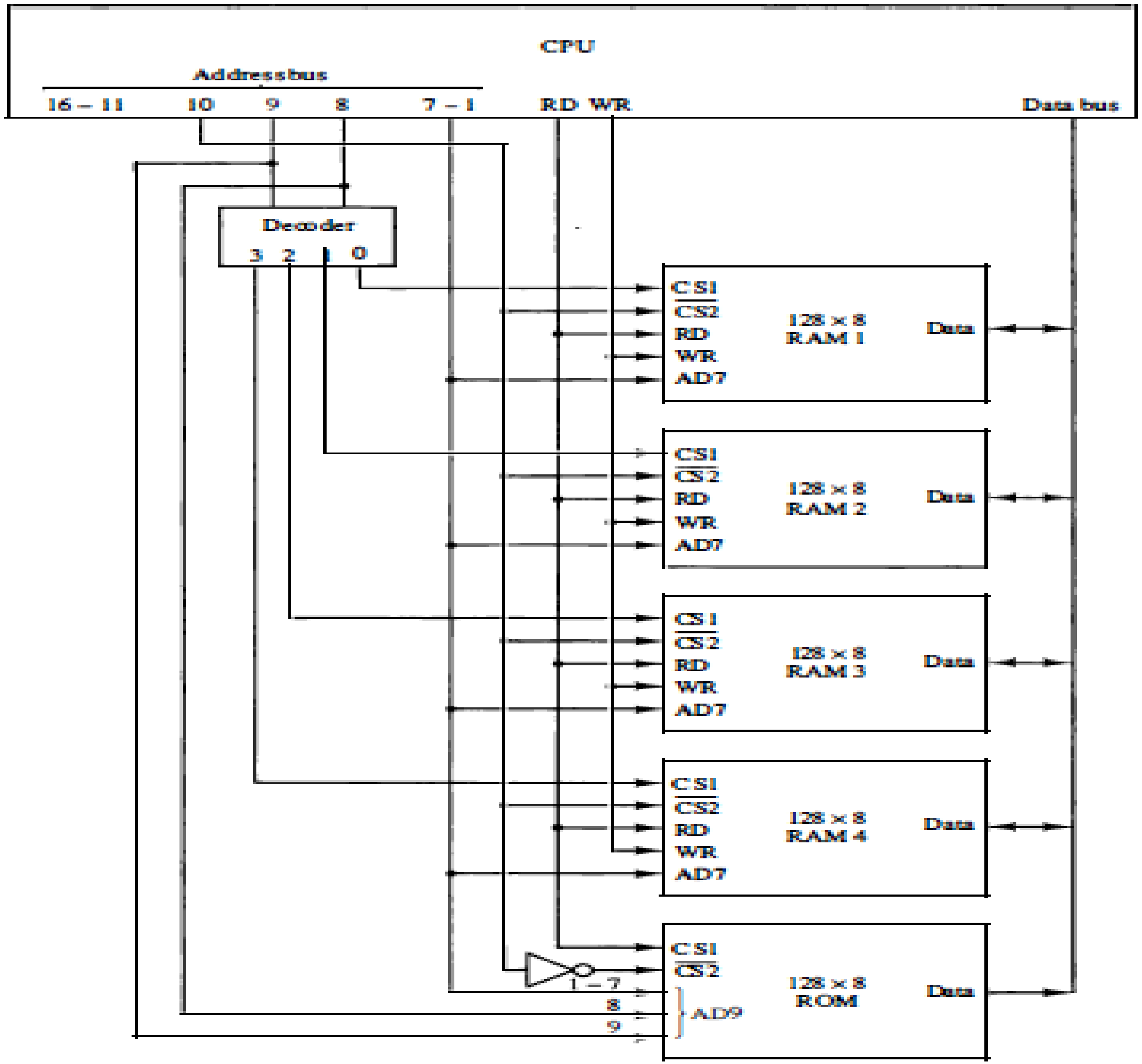


Figure 12-3 Typical ROM chip.

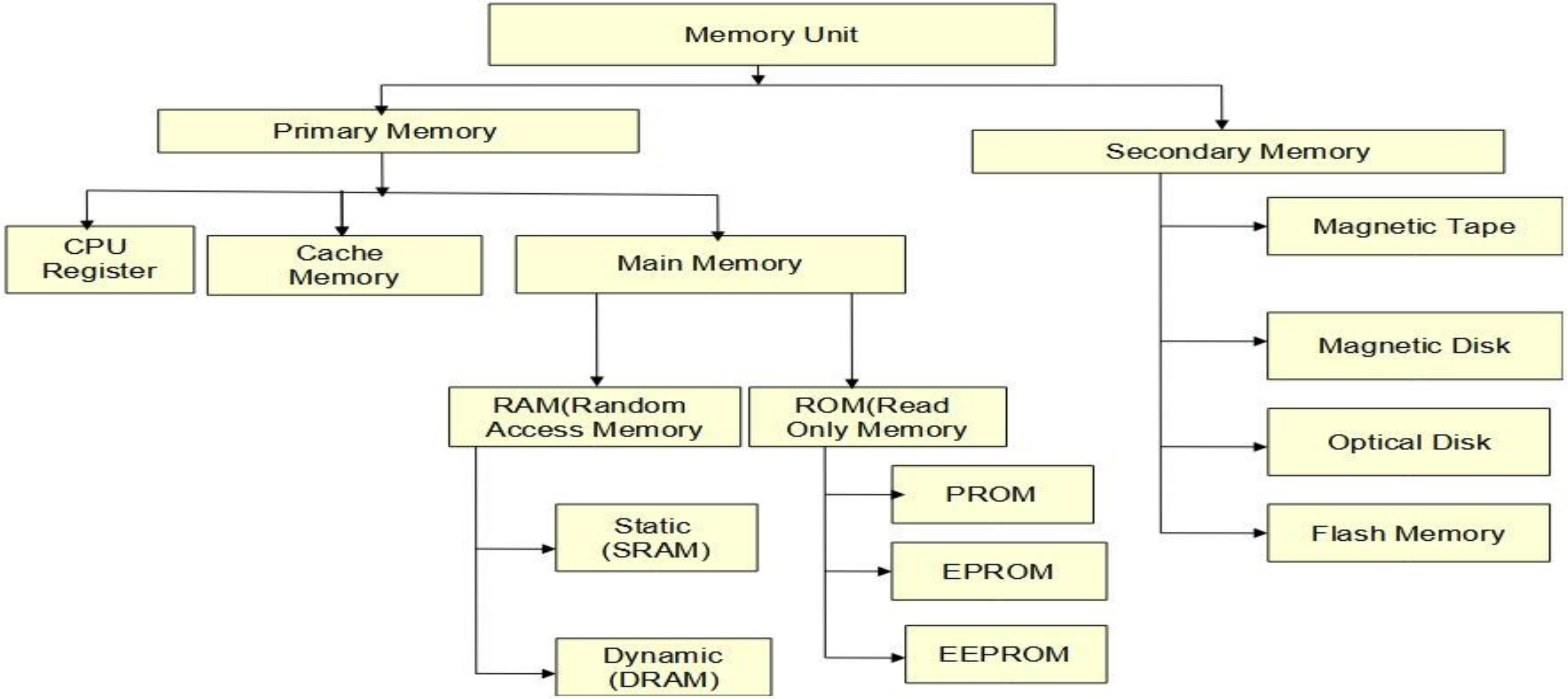
MEMORY CONNECTION TO CPU

- RAM and ROM chips are connected to a CPU through the **data and address buses.**
- Each RAM receives the seven low-order bits of the address bus to select one of 128 possible bytes.
- when address lines 8 and 9 are equal to 00, the first RAM chip is selected. When 01, the second RAM chip is selected, and so on.
- The particular RAM chip selected is determined from lines 8 and 9 in the address bus.
- The selection between RAM and ROM is achieved through bus line 10. **The RAMs are selected when the bit in this line is 0, and the ROM when the bit is 1.**
- the ROM chip to be enabled only during a read operation.
- Address bus lines 1 to 9 are applied to the input address **of ROM without going through the decoder.**
- **The data bus of the ROM has only an output capability, whereas the data bus connected to the RAMs can transfer information in both directions** .



Auxiliary Memory

- The most common auxiliary memory devices used in computer systems are magnetic disks and tapes.
- It Provides Permanent data storage of device and it has also less speed of data transfer.



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Auxiliary Memory

Magnetic Tape

- The tape itself is a strip of plastic coated with a magnetic recording medium.
- When we required data read or write that **magnetic material helps to data captured**.
- In magnetic tape only one side of the ribbon is used for storing data.
- It is **sequential memory** which contains thin plastic ribbon to store data and coated by magnetic oxide.
- Data read/write access speed is slower because of sequential access.
- Generally taps are used for **backup storage only.**
- It can remove a specific data and save another data at the same place. **Therefore it can be reused.**
- Magnetic material is coated only one side **when it is damage then** can not easily read or write.
- Magnetic tapes are highly vulnerable to **damage from dust or careless handling.**

Auxiliary Memory

Magnetic Disks

- **A magnetic Disk** is a type of secondary memory that is a **flat disc covered with a magnetic coating to hold information.**
- Bits are stored on the magnetized surface in circles **called tracks.**
- The tracks are commonly divided into **sections called sectors**.
- Disk is divided into 2 surfaces 1) Upper surface 2) Down surface
- Both surface to capable read the data and also write the data.
- Both sides of the disk are used.
- **Several disks may be stacked on one spindle with read/write heads available on each surface.**

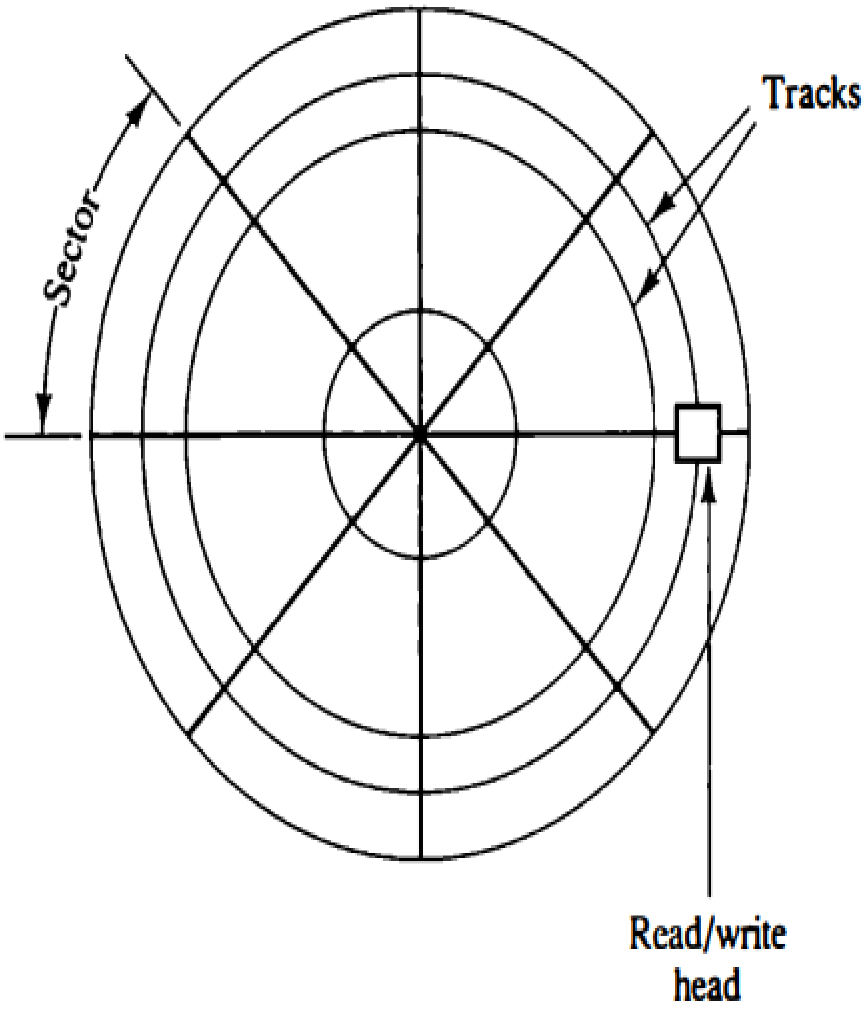


Figure 12-5 Magnetic disk.

Auxiliary Memory

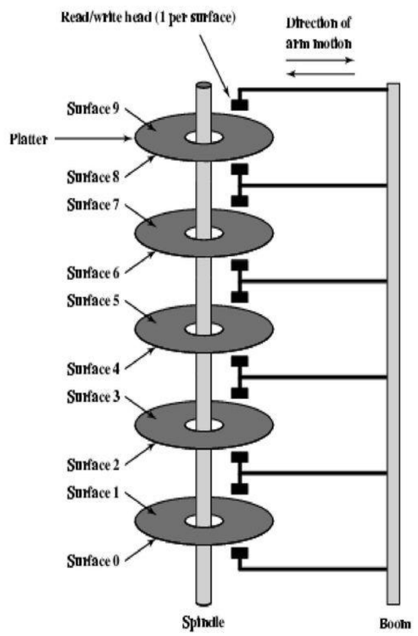
Data can be modified or can be deleted easily in the magnetic disk.
It also allows **random access to data**. Because of the concept of sectors.

- **Using Magnetism** read write heads can read/write data on to the disk.
- If magnetism power is less then you can not write in Proper manner.
- **To overcome** this we have CD/DVD, it is Optical disk (using light ray we can read/write the data, just mechanism is changed)

Access time – − Data access time is the actual time needed to access the data.

Data transfer time − Data transfer time is the actual time needed to send the data.

Multiple Platters



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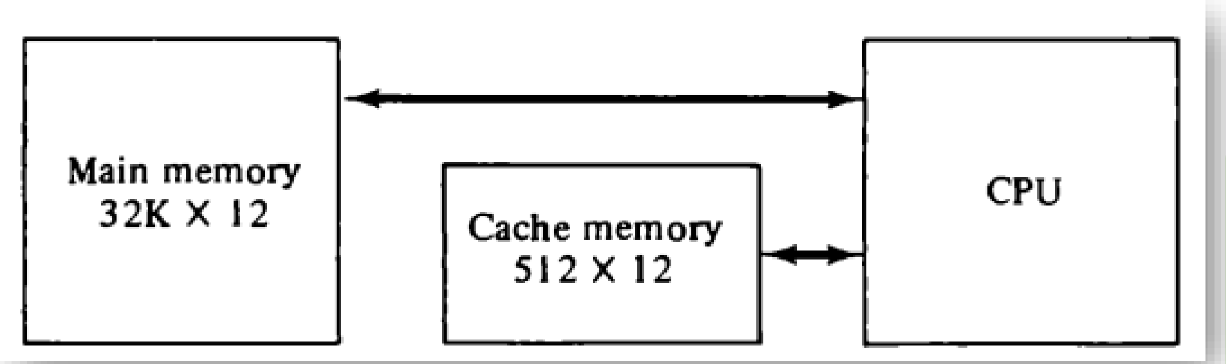
Cache Memory

Cache Memory and Types of memory mapping in cache memory

- Cache is a fast small capacity memory that should hold those information which are most likely to be accessed.
- If the word is found in the cache, it is read from the fast memory. If the word addressed by the CPU is not found in the cache, the main memory is accessed to read the word.
- **The transformation of data from main memory to cache memory is referred to as a mapping process.**



- Three types of mapping procedure :
  - 1. Associative mapping
  - 2. Direct mapping
  - 3. Set-associative mapping

- NOTE:
- The performance of cache memory is frequently measured in terms of a quantity called **hit ratio.**
- When the CPU refers to memory and finds the word in cache, it is said to produce a **hit.**
- If the word is not found in cache, it is in main memory and it counts as a **miss.**

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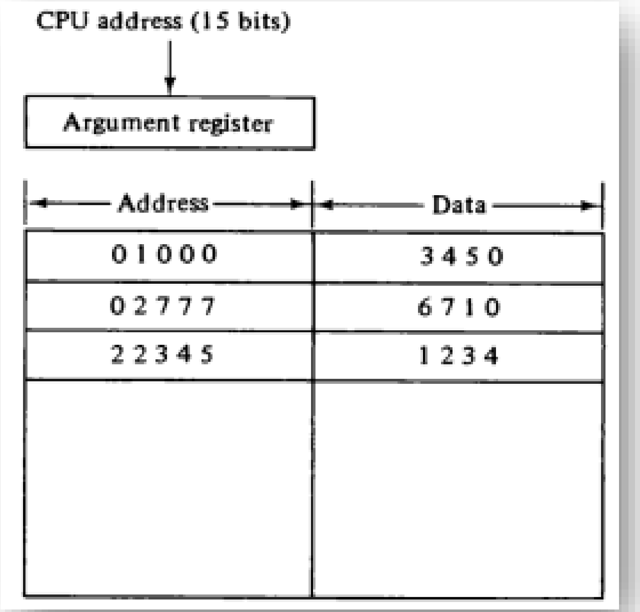
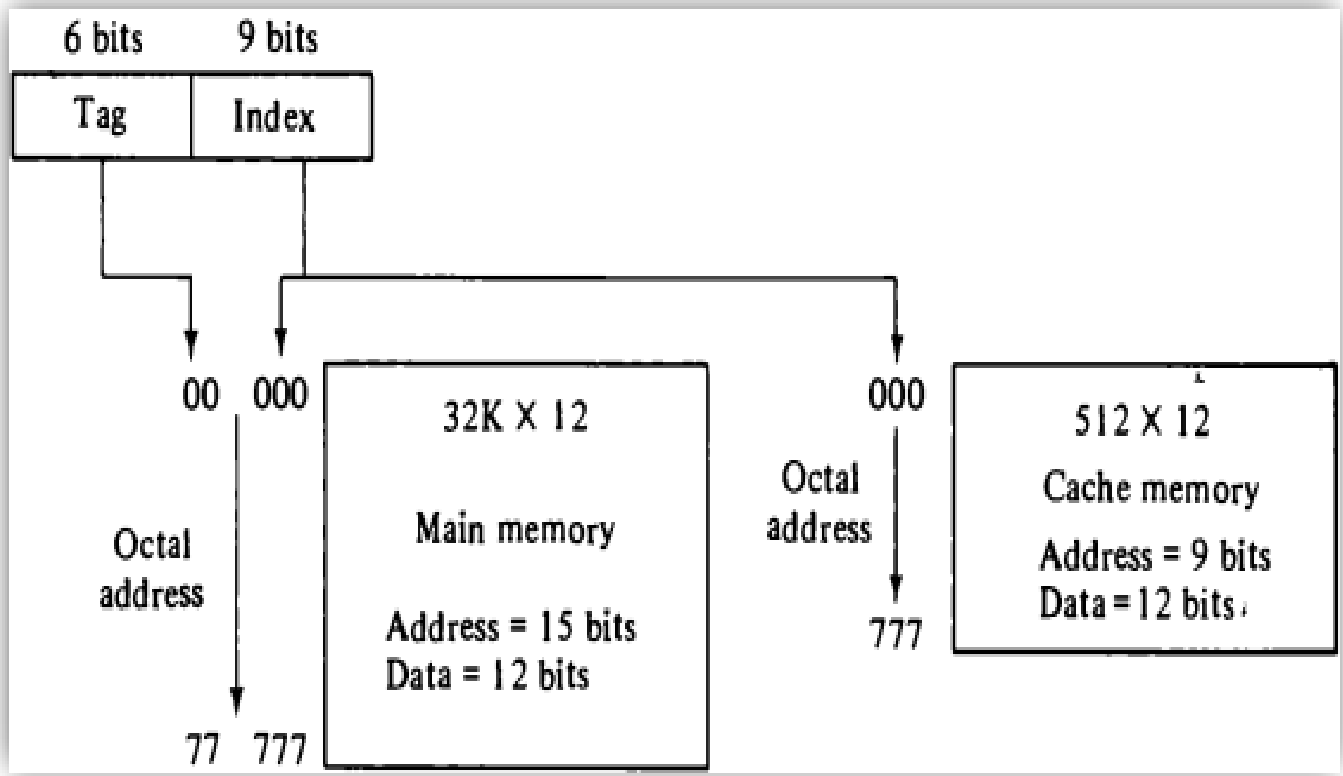
- **1) Associative mapping:**
- Consider the main memory can store 32K words of 12 bits each.
- The cache is capable of storing 512 of these words at any given time. For every word stored in cache, there is a duplicate copy in main memory.
- The CPU communicates with both memories. It first sends a 15-bit address (for 32 KB) to cache.
- **If there is a hit** the CPU accepts the 12-bit data from cache, **if there is** miss, the CPU reads the word from main memory and the word is then transferred to cache.



- Fig. shows three words presently stored in the cache. **The address value of 15 bits is shown as a five-digit octal number** and its corresponding **12-bit word is shown as a four-digit octal number.**
- A CPU address of 15 bits is placed in the argument register and the associative memory is searched for a matching address. If the address is found the corresponding 12-bit data is read and sent to CPU.
- If no match occurs, the main memory is accessed for the word. The address data pairs then transferred to the associative cache memory.

❖ if cache memory have empty space then it can directly transfer but if there is no space so it can replace the data and that FIFO replacement policy is used.

- 2) **Direct mapping:**



- The CPU address of 15 bits is divided into two fields.
- The nine least significant bits constitute the index field and the remaining six bits from the tag field.
- The number of bits in the index field is equal to the number of address bits required to access the cache memory.

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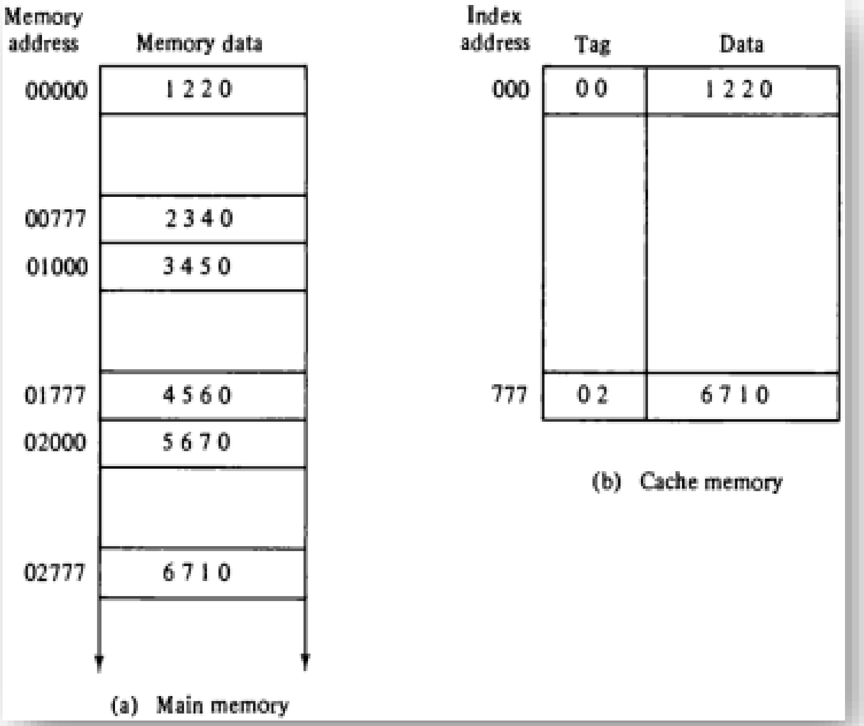
- Fig. shows the internal organization of the words in the cache memory.
- Each word in cache consists of the data word and its associated tag.
- When the CPU generates a memory request the index field is used for the address to access the cache.
- The tag field of the CPU address is compared with the tag in the word read from the cache. If the two tags match, there is a hit and the desired data word is in cache.



- If there is no match, there is a miss and the required word is read from main memory. It is then stored in the cache together with the new tag, replacing the previous value.
- The word at address zero is presently stored in the cache (index = 000, tag = 00, data = 1220).
- Suppose that the CPU now wants to access the word at address 02000. The index address is 000, so it is used to access the cache. The two tags are then compared. The cache tag is 00 but the address tag is 02, which does not produce a match. Therefore, the main memory is accessed and the data word 5670 is transferred to the CPU.
- The cache word at index address 000 is then replaced with a tag of 02 and data of 5670.
- The disadvantage of direct mapping is that two words with the same index in their address but with different tag values cannot reside in cache memory at the same time.

Cont...

3) Set – associative mapping

- In these each word of cache can store two or more words of memory under the same index address.
- Each data word is stored together with its tag and the number of tag-data items in one word of cache is said to form a set.
- Each index address refers to two data words and their associated terms.
- Each tag required six bits and each data word has 12 bits, so the word length is 2 (6+12) = 36 bits.
- An index address of nine bits can accommodate 512 words.
- Thus the size of cache memory is 512 x 36. It can accommodate 1024 words or main memory since each word of cache contains two data words.
- The words stored at addresses 01000 and 02000 of main memory are stored in cache memory at index address 000.

Index	Tag	Data	Tag	Data
000	01	3450	02	5670
777	02	6710	00	2340

3) Set – associative mapping

- Similarly, the words at addresses 02777 and 00777 are stored in cache at index address 777.
- When the CPU generates a memory request, the index value of the address is used to access the cache.
- The tag field of the CPU address is then compared with both tags in the cache to determine if a match occurs.
- When a miss occurs in a set-associative cache and the set is full, it is necessary to replace one of the tag-data items with a new value.

Index	Tag	Data	Tag	Data
000	0 1	3 4 5 0	0 2	5 6 7 0
777	0 2	6 7 1 0	0 0	2 3 4 0

Associative Memory(content addressable memory(CAM)

- Generally we can access data using their address , that doesn't matter the memory is random access memory or sequential access memory.

- **The time required to find an item stored in memory can be reduced using we can direct access stored data itself rather than by an address.**

- **A memory unit accessed by content is called an associative memory or content addressable memory (CAM).**

- **This type of memory is accessed simultaneously and in parallel on the basis of data content rather than by specific address.**

- When a word in written in an associative memory no address is given.

- The memory is also capable of finding an empty unused location to store the word(Data).

Associative Memory(content addressable memory (CAM

- **When a word is to be read from an associative memory, the content of the word, or part of the word, is specified. (Eg, search data 11100011 then you can Provide 11100011 or 111)**

- **The memory locates all words which match the specified content and marks them for reading.**

- **An associative memory is more expensive than a random access memory because each cell must have storage capability as well as logic circuits for matching its content with an external argument(Content).**

- **For this reason, associative memories are used in applications where the search time is very critical and must be very short (for example google you can only give content)**

Hardware organization of Associative Memory

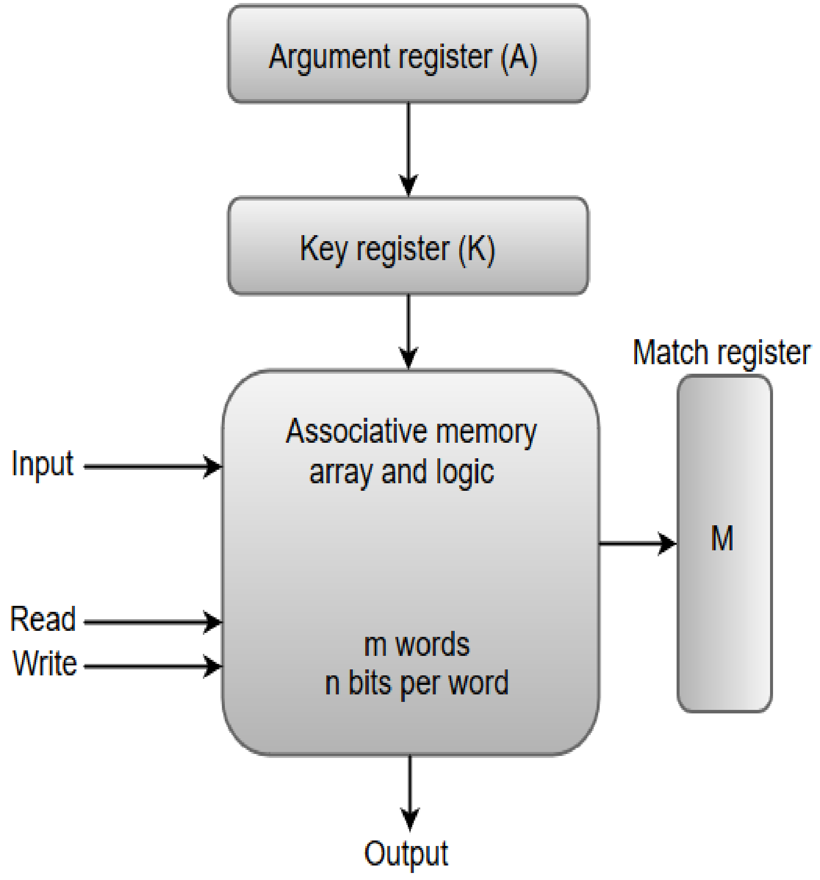
Argument Register – it contains word to be searched.

Key Register – It specify which Part of the word to be Compared with Word in memory. (Supposed you can compared initial 3 bits or else also compare entire word)
If all bits in key register are 1's, the entire word should be compare, otherwise only the bits having 1's in their corresponding positions are compared.

Associative memory - it contains the word that to be compared with the argument word .

Match Register – bits are matched with argument register bits to Associative memory then set to 1 in match register (M bit) or match is not occurred set to 0 in M bit.

After the matching bits set 1's ,we can read the data and Reading is sequential manner .



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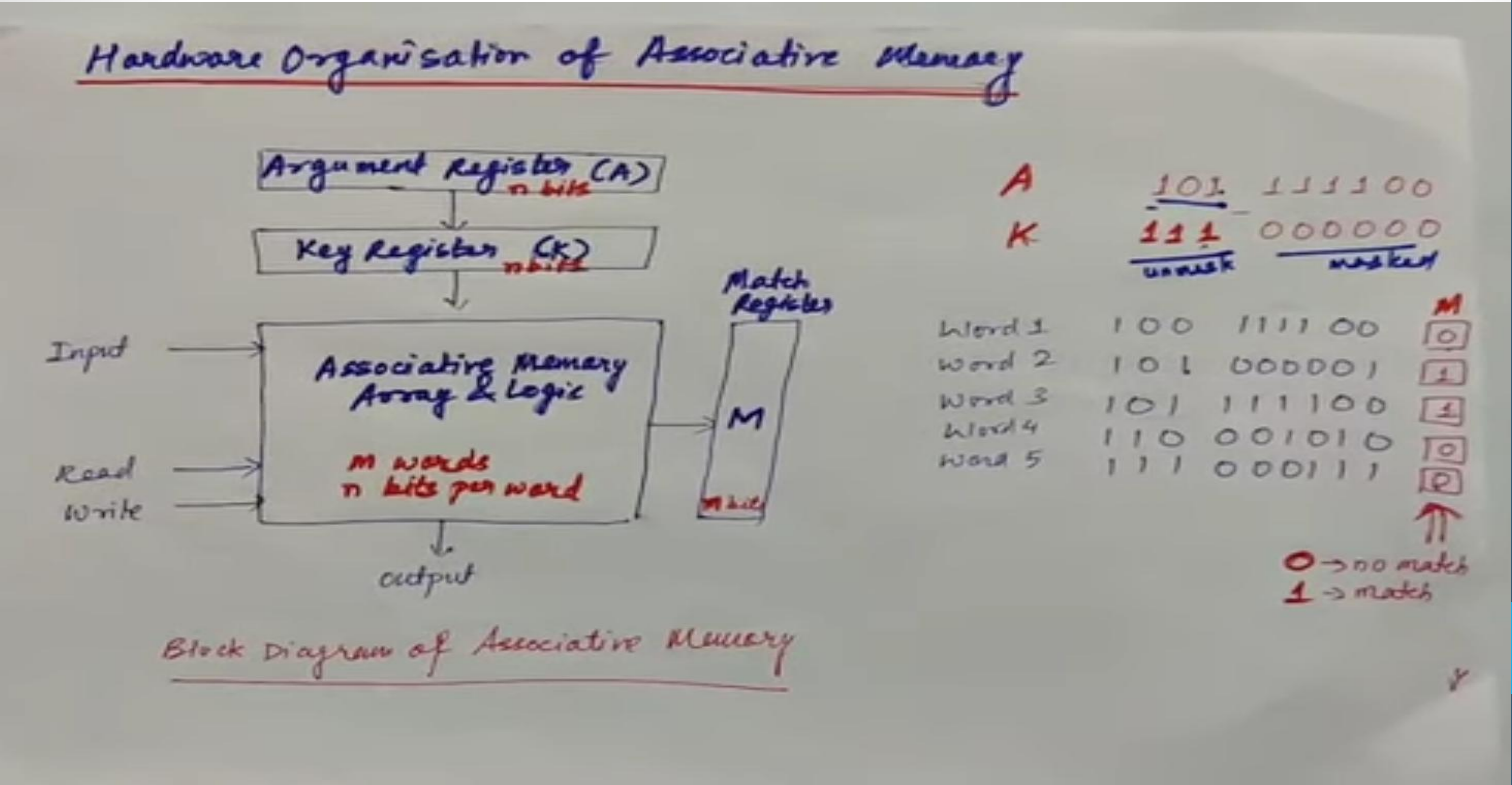
Example


Hardware Organisation of Associative Memory
Argument Register (A)
n bits
Key Register (K)
n bits
Match Register
Input
Associative Memory Array & Logic
m words
n bits per word
Read
Write
M
m bits
output
Block Diagram of Associative Memory
A 101 111100
K 111 000000
unmask
masked
M
Word 1 100 111100 0
Word 2 101 000001 1
Word 3 101 111100 1
Word 4 110 001010 0
Word 5 111 000111 0
0 → no match
1 → match

Virtual Memory

- **Virtual Memory** – it is technique allows users to use more memory for a Program than the real memory of a computer.
- Virtual memory is the concept that gives **illusion to the user that they will have main memory equal to the capacity** of secondary storage media (auxiliary memory).

- **Need of memory :-**
- **Virtual memory is a imaginary memory we are assuming if we have a Program that exceeds the size of main memory than we need to use the concept of virtual memory.**
- **For example – (Main memory size 4GB and Program size 16GB so now currently used Program transfer to the Main memory in some Portion then inactive part of Main memory also transfer to the auxiliary memory this Process called Swapping.)**
- **Concept of virtual memory only keeps those Programs we currently used in main memory and rest of part into the hard disk.**

Virtual Memory

- How the Virtual Memory Is implemented .

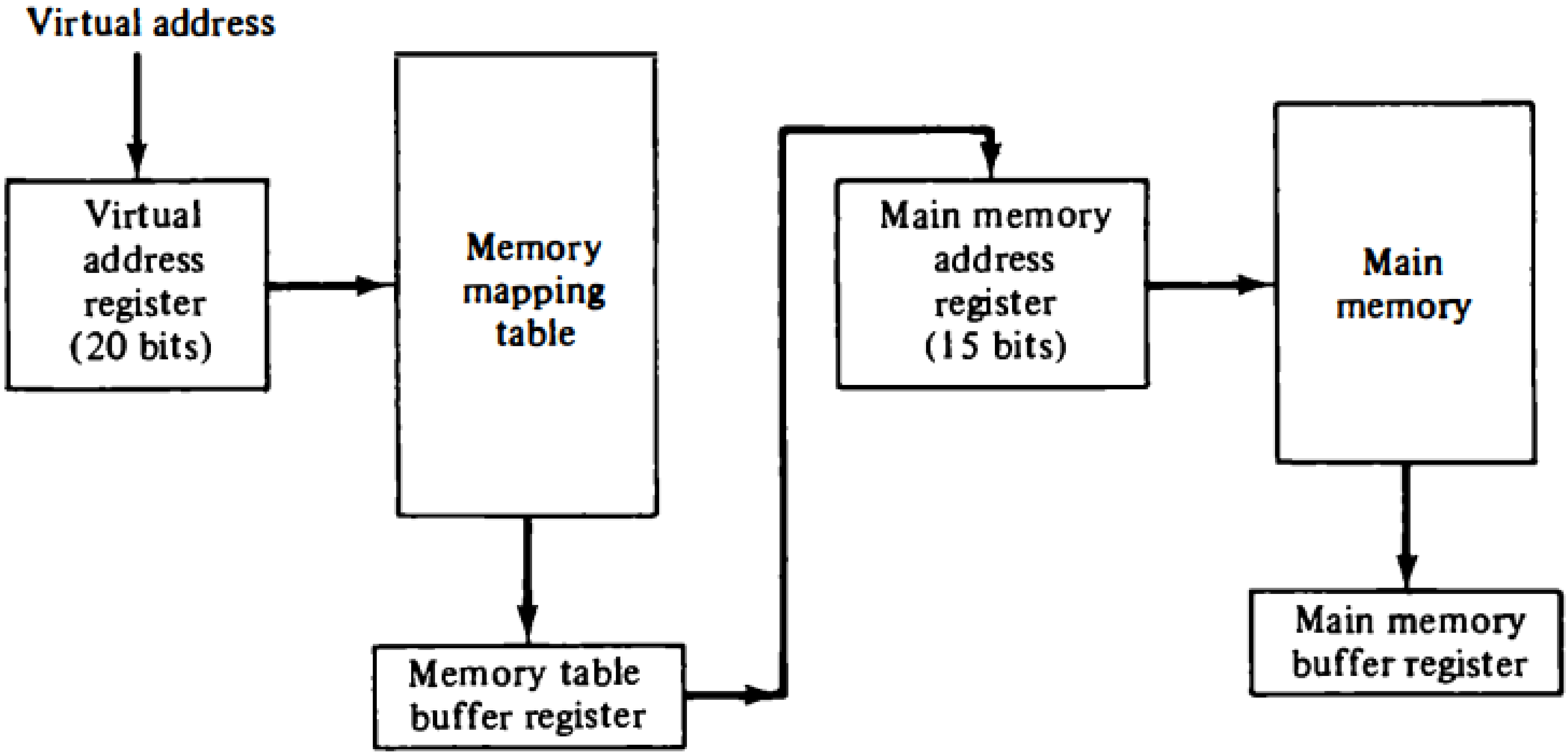
- To implement VM, a portion of Hard Disk is Reserved by system.
- The extra Memory does not actually exist in RAM, but it is the storage space on the hard disk which is implemented with the help of swapping between main memory and hard disk.

- **Virtual addresses** – address Programmers are used.(Logical address)
- Set of such addresses called **address space.**
- An address in main memory is called **Physical addresses.**
- Set of such addresses called **Memory space.**

Virtual Memory

- **The address translating or mapping** is handled automatically by the h/w by means of a Mapping Table.

Figure 12-17 Memory table for mapping a virtual address.



Virtual Memory

- **Two Virtual Memory Management Techniques, which does the mapping from virtual address to Physical address :-**

1) Paging (demand Paging)

2) segmentation (demand segmentation)

- **Address mapping using Pages**
- In paging address space (virtual memory) and Memory space (Physical memory) is divided into equal size groups.
- address space (virtual memory) is divided into **equal size pages.**
- Memory space (Physical memory) is divides into **equal size Blocks**. (Page frame)
- Page size = Block size

Virtual Memory

- **Consider a Computer with an address space = 8k and a memory space = 4k**
- If we split each **equal size groups of 1K words** , we obtain 8 pages and 4 blocks,
- It means at any given time up to 4 pages of address space may resides in main memory in any one of the 4 blocks.
- **Virtual address = 13 bits**
- **Physical address = 12 bits**

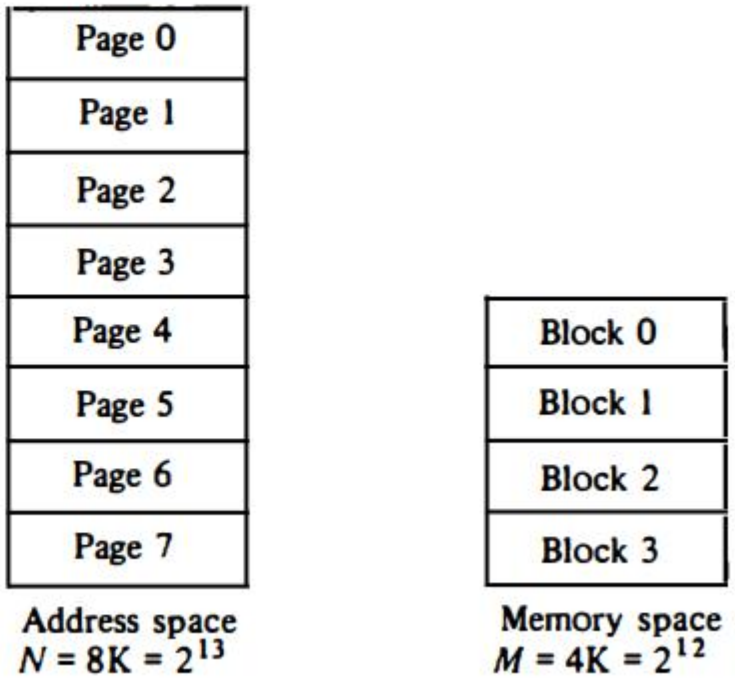


Figure 12-18 Address space and memory space split into groups of 1K words.

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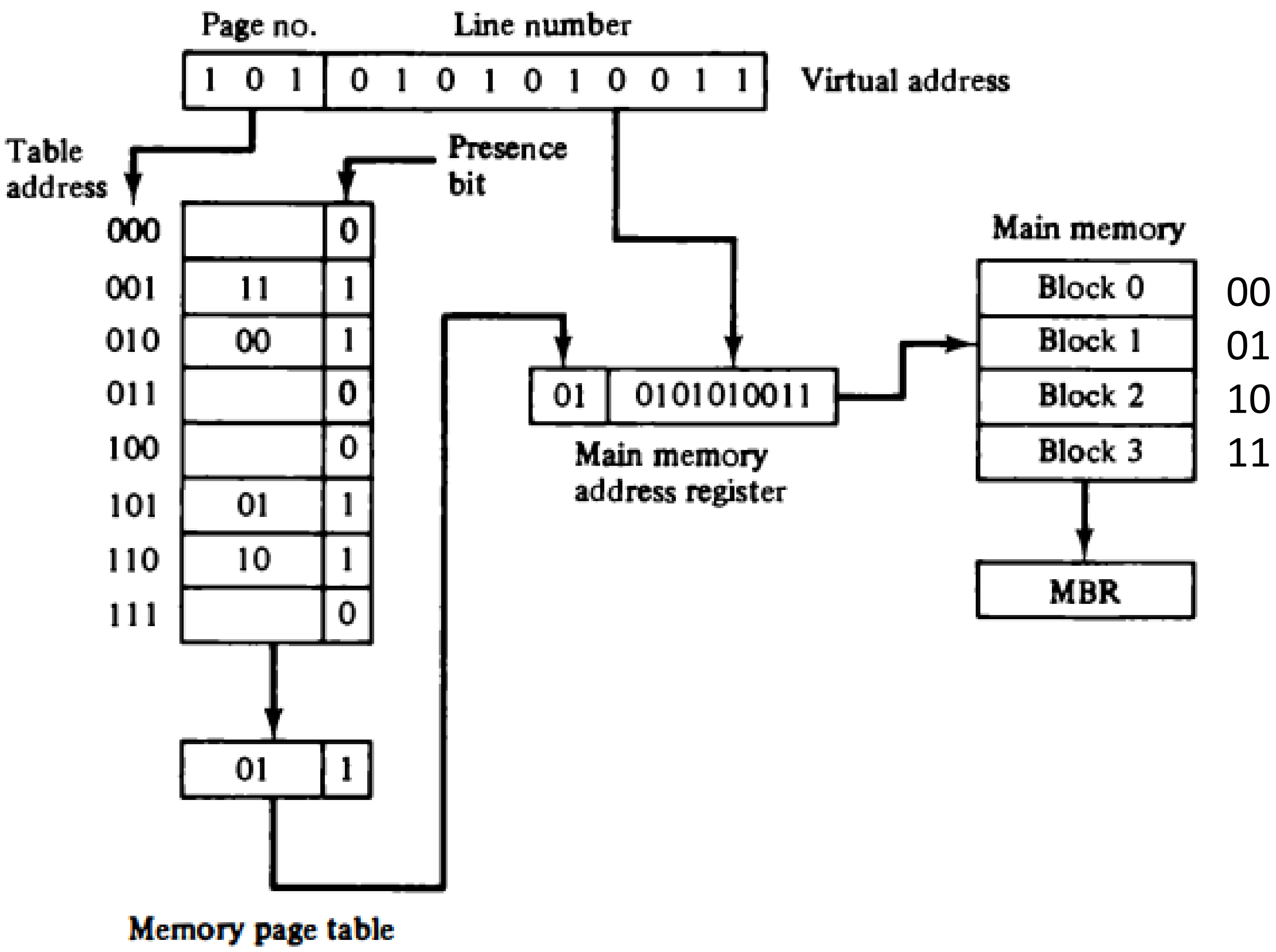
Virtual Memory

- Page system , the virtual address divided into two parts :-
- 1) Page number
- 2) Line number
- High order bits are called the page number
- Low order bits are called the line number.

- Cpu access which page their not present in your main memory then **occur page fault** then cpu replace one block and also that called page replacement policy.
- (FIFO, list recently used , Optimal page replacement)

Figure 12-19 Memory table in a paged system.



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DRAM	SRAM
1. Constructed of tiny capacitors that leak electricity.	1.Constructed of circuits similar to D flip-flops.
2.Requires a recharge every few milliseconds to maintain its data.	2.Holds its contents as long as power is available.
3.Inexpensive.	3.Expensive.
4. Slower than SRAM.	4. Faster than DRAM.
5. Can store many bits per chip.	5. Can not store many bits per chip.
6. Uses less power.	6.Uses more power.
7.Generates less heat.	7.Generates more heat.
8. Used for main memory.	8. Used for cache.

QUESTIONS ASKED IN GTU EXAM

Q.1 Write a detailed note on associative memory.

Q.2 Write a brief note on memory hierarchy.

Q.3 Explain any two types of mapping procedures when considering the organization of cache memory

Q.4 Draw and criticize memory hierarchy in a computer system.

Q.5 What is RAM and ROM?

Q.6 Write a short note on virtual memory.

Q.7 What is cache memory address mapping? Compare and contrast direct address mapping and set-associative address mapping.

Q.8 What is the importance of virtual memory?

Q.9 Elaborate content addressable memory (CAM).

Q.10 Explain memory hierarchy in brief.

Q.11 What do you mean by cache memory? Justify the need of cache memory in computer systems.

Q.12 What is RAM and ROM?

Q.13 What is cache memory address mapping? Which are the different memory mapping techniques? Explain any one of them in detail.

Q.14 What is associative memory? Explain